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Started	Wednesday, 21 October 2026, 10:45 PM
Completed	Wednesday, 21 October 2026, 10:52 PM
Duration	6 mins 58 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

- a. Dividing the list into halves
- b. Bubbling up the largest element to its correct position
- c. Bubbling up the smallest element
- d. Sorting the entire list in one pass

The correct answer is: Bubbling up the largest element to its correct position

Question **2**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

- a. Arranging data in a particular format
- b. Inserting data into a list
- c. Deleting data from a list
- d. Searching for data in a list

The correct answer is: Arranging data in a particular format

Question **3**

Complete

Mark 0.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

- a. Bubble Sort
- b. Linear Search
- c. Merge Sort
- d. Quick Sort

The correct answer is: Merge Sort

Question **4**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

a. arrange()
b. order()
c. sorted()
d. sort()

The correct answer is: sorted()

Question **5**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

a. By sorting data in a single pass
b. By comparing elements sequentially
c. By breaking the input into smaller parts and merging them
d. By using the bubble-up method

The correct answer is: By breaking the input into smaller parts and merging them

Question **6**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

a. It makes the search operation slower
b. It allows for faster searching
c. It increases the number of comparisons needed
d. It has no effect on the search operation

The correct answer is: It allows for faster searching

Question **7**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

a. It repeatedly finds the minimum element and moves it to the sorted part of the list
b. It builds a sorted array one element at a time
c. It compares adjacent elements and swaps them if necessary
d. It divides the list into two halves, sorts each half, and then merges them

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **8**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

a. Bubble Sort
b. Merge Sort
c. Binary Search
d. Linear Search

The correct answer is: Merge Sort

Question **9**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

a. 40,70,80,90,120,150,200,300,400,470
b. 40,80,90,150,200,300,400,470,70,120
c. 200,470,80,150,40,90,300,400,70,120
d. 80,150,200,470,40,90,300,400,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **10**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

a. Rearranging data without sorting
b. Sorting data sequentially
c. Sorting data in a single pass
d. Dividing the input into parts, solving each part, and combining the solutions

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **11**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

a. All of the above
b. Selection
c. Duplicates
d. Searching

The correct answer is: All of the above

Question **12**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

a. It makes the data unsorted
b. It complicates the selection of items
c. It makes it easier to select items based on their relationship to the rest of the items
d. It slows down the process

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **13**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

- a. Provides a basis for other algorithms to work efficiently
- b. Makes code execution slower
- c. Makes data harder to manage
- d. Decreases the efficiency of algorithms

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **14**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

- a. Insertion Sort
- b. Quick Sort
- c. Merge Sort
- d. Bubble Sort

The correct answer is: Merge Sort

Question **15**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

- a. Linear Search
- b. Bubble Sort
- c. Merge Sort
- d. Binary Search

The correct answer is: Bubble Sort